

TED (15) - 6093

Reg.No.....

REVISION-2015

Signature.....

MODEL QUESTION PAPER
DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/
MANAGEMENT/COMMERCIAL PRACTICE
TYRE TECHNOLOGY

(Time : 3 hours)

(Maximum marks : 100)

PART – A

Marks

I Answer all questions in one or two sentences. Each question carries 2 marks

1. Define pneumatic tyre,
2. Distinguish between bias and radial tyre.
3. Describe PCI.
- 4 .List laboratory tests for tyre cords..
5. Advantages of hot retreading process

(5x2=10)

PART – B

II Answer **any five** of the following questions. Each question carries 6 marks

1. Explain production of sidewall with a suitable formulation.
2. Describe the basic tyre constructions.
3. Explain the curing of cycle tyre
4. Describe the methods of determination of cord strength and casing strength.
5. Describe the auxiliary materials.
6. Explain the preparation of inner liner of the tubless tyre.
7. Describe the production of solid tyre with suitable formulation

(5x6=30)

PART- C

(Answer one full question from each unit. Each full question carries 15 marks)

MODULE-I

- III (a) With a neat sketch explain the functions of different components of a bias tyre. (8)
- (b) List out the major tyre manufacturing industries in India and abroad. (7)

OR

- IV (a) Distinguish between bias and radial tyre. (8)
- (b) Write the functions pneumatic tyre. (7)

MODULE-II

- V (a) Describe the need of adhesive treatment for tyre cords and requirement of adhesive solution.. (8)
- (b) Describe adhesive treatment for rayon, nylon, and polyester. (7)

OR

- VI (a) Explain the curing of bias tyre. (8)
- (b) Design suitable compounds for sidewall of pneumatic tyre (7)
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MODULE-III

VII (a) Explain the monoband and drum building method for cycle tyre. (8)

(b) With a suitable formulation explain the production of cycle tyre tread (7)

OR

VIII (a) Design typical compound for truck tyre re- tread and justify your selection of ingredients (8)

(b) Explain the production of cushion gum for tyre re-treading with suitable formulation (7)

MODULE-IV

IX (a) Explain laboratory testing of tyre cords. (8)

(b) Differentiate destructive non-destructive test for tyre. and explain the non-destructive tests. (7)

OR

X (a) Explain destructive and nondestructive tests done on automobile inner tube.. (8)

(b) Explain the cut tyre analysis. (7)

